$\frac{1}{\sin^2 x} = \csc^2 x$

$\int \csc^2 x \, dx$

Quiz 8:-

C6:- (A)

$$\int x \csc^2 x \, dx$$

$u = x$ | $dv = \csc^2 x \, dx$

$du = dx$ | $v =$

(B) $$\int \frac{3x}{e^x} \, dx$$

Let $u =$ | $dv = e^{-x} \, dx$

$du =$ | $v =$

$$\int u \cdot dv = uv - \int v \, du$$

$$= e^{-x} \frac{3}{2} x^2 - \int \frac{3}{2} (x^2 e^{-4}) \, dx$$

CT: $\int \cos^3 x \sin^2 x \, dx = \int \cos x \sin^2 x \cos^2 x \, dx$

$= \int \cos x (1 - \cos^2 x) \cos^2 x \, dx$

let $u = \sin x$
$du = \cos x \, dx$

$= \int (\cos x - \cos^3 x) \cos^2 x \, dx$

$= \int (\cos^3 x - \cos^5 x) \, dx$

$= \int \cos^3 x \, dx - \int \cos^5 x \, dx$

$= \int \cos^2 x \cos x \, dx - \int (\cos^2 x)^2 (\cos x) \, dx$

$= \int (1 - \sin^2 x) \cos x \, dx - \int (1 - \sin^2 x)^2 \cos x \, dx$

$$= \int \cos x\,dx - \int \sin^2 x \cos x\,dx$$

$$- \int (1 - 2\sin^2 x + \sin^4 x)\cos x\,dx$$

$$= \sin x - \int u^2\,du - \int (\cos x - 2\sin^2 x \cos x + \sin^4 x \cos x)\,dx$$

Let $u = \sin x$, $du = \cos x\,dx$

$$= \sin x - \frac{\sin^3 x}{3} - \int \cos x\,dx + \int 2\sin^2 x \cos x\,dx - \int \sin^4 x \cos x\,dx$$

$$= \sin x - \frac{\sin^3 x}{3} - \sin x + \frac{2\sin^3 x}{3} - \frac{\sin^5 x}{5} + C$$

$$= C - \frac{\sin^3 x}{3} + \frac{2\sin^3 x}{3} - \frac{\sin^5 x}{5}$$

$$= \frac{\sin^3 x}{3} - \frac{\sin^5 x}{5} + C$$

$$\int \cos^3 x \sin^2 x\,dx = \frac{\sin^3 x}{3} - \frac{\sin^5 x}{5} + C$$